Paul Chukwu

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Professional Summary

Computer Engineering student specializing in hardware design, **SoC design, embedded systems, FPGA** development, digital logic, and system-level engineering. Skilled in **C, C++, Python, VHDL, SystemVerilog,** Assembly, CAD modelling, microcontrollers, and telemetry. **Completed 20+ FPGA** projects including UART systems, video timing pipelines, and hardware-software integrations.

Education

BASc in Computer Engineering, uOttawa (Expected 06/2026)

- Term Dean's List, Honor Roll, UOSU Scholarship for Innovation in Sustainability (2024)
- Relevant coursework: FPGA, Quartus-II Pro, Microcontrollers, VHDL/System Verilog, Circuit Design and Testing.

Skills

- Languages: C, C++, Python, Java, Kotlin, System Verilog, VHDL, Assembly, HTML-CSS, SQL.
- Tools: Android Studio, VS Code, Web flow, Quartus-II Pro, FPGA, Microcontrollers, CAD.
- GitHub/GitLab Systems: Operating Systems, Virtual Machines, Hypervisors, Embedded Systems Other: Debugging with GDB, scripting in Python, QEMU (emulator).

Projects

- **Soil/Water Quality System – (EWB-Sahel):** Built soil moisture and water level detection using sensors and an LCD1602 (see demo videos on my website).
- **Embedded UART + CLI Games – QNX Hackathon:** Designed UART comms + interactive CLI game.
- **Eco Sorter – Smart Recycling Device (1st Place)** –Built automated recycling system both hardware and software integration using **Python/Java/C.** Then built actual prototype, with different sensors.
- **UART Communication**: Built structural UART transmitter/receiver in **VHDL+ Verilog**, integrating FSM and ASM concepts for embedded systems. Created TDR, TSR, SCSR, and controller logic.
- **Telemetry System – Shell Eco Marathon**: Developed speedometer and button reactor in C/C++ using interrupts and time delays. Implemented magnet-based speedometer system.
- **Ross Video Hackathon:** Designed **1080x720 FPGA** video display system in **System Verilog**, mastering video timing and low-level design. Created animations, shapes, and text render logic.
- **Mirror Cap Assistive Hardware:** Built ultrasonic/LDR/servo feedback device.
- **Chess Robotic Arm:** Designing a Robotic arm that simulates a board where you can play chess, using Arduino IDE, C, Python(Training the AI).

Work Experience

Intern, Snappy Techs (08/2021 - 01/2023)

- Assisted staff in technical projects, gaining hands-on experience in software and hardware.
- Tested numerous devices for errors. Including overstressing in order to test durability.

Achievements

- 1st place: Eco Sorter (Self-Recyclable Bin) | 2nd place: Hack the Hill FPGA Project
- IEEE Member | Six Sigma, CPR/AED, First Aid Certified